E-Commerce Order Management Database System

Week 1 – Understanding the Business Problem

1. Introduction

The **E-Commerce Order Management Database System** is designed to manage the day-to-day operations of an online shopping platform similar to Amazon or Flipkart. The system helps customers browse products, place orders, make payments, and track deliveries. It also enables administrators to manage products, inventory, suppliers, customers, and shipments through a centralized database. This system ensures efficient data management, reduces redundancy, and improves accuracy.

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2. Business Problem

Traditional methods of managing customer information, product inventory, orders, and payments using spreadsheets or manual records are time-consuming and prone to errors. As the number of customers and products increases, it becomes difficult to maintain accurate records and provide fast service.

An automated database system is required to:

- Store customer and product information securely.
- Process customer orders efficiently.
- Manage inventory automatically.
- Track payments and shipments.
- Generate reports for business analysis.

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3. Major Activities in an E-Commerce System

Customer Registration

- Customers create an account using their personal details.
- The system stores customer information securely.
- Registered users can log in and manage their profiles.

Product Browsing

- Customers browse products by category.
- Search products using keywords.
- View product details such as price, description, and stock availability.

Order Placement

- Customers add products to the shopping cart.

- Place orders after confirming the selected products.
- The system generates an order ID and records order details.

Payment Processing

- Customers make payments using online payment methods.
- The system records payment details and payment status.
- Successful payments are linked with customer orders.

Shipment Tracking

- Orders are prepared for delivery after payment confirmation.
- Shipment details are recorded.
- Customers can track the delivery status until the order is delivered.

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4. Functional Requirements

The proposed system should provide the following functionalities:

Customer Management

- Register new customers.
- Store customer details.
- Update customer profiles.

Product Management

- Add new products.
- Update product details.
- Maintain product categories.
- Manage inventory stock.

Order Management

- Place customer orders.
- Store order information.
- Update order status.
- Maintain order history.

Payment Management

- Record payment transactions.
- Maintain payment status.
- Support multiple payment methods.

Shipment Management

- Record shipment information.
- Track delivery status.
- Maintain delivery addresses.

Reporting

- Generate sales reports.
- Generate customer purchase history.
- Generate product-wise revenue reports.

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5. Scope of the System

The system includes the following modules:

- Customer Management
- Product Management
- Category Management
- Supplier Management
- Order Management
- Payment Management
- Shipment Management
- Review Management
- Business Reporting

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6. Expected Benefits

- Faster order processing.
- Accurate inventory management.
- Secure customer and payment records.
- Reduced manual work.
- Improved customer satisfaction.
- Easy generation of business reports.
- Better decision-making through organized data.

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7. Conclusion

The **E-Commerce Order Management Database System** provides a complete solution for managing online shopping operations. It integrates customer registration, product browsing, order placement, payment processing, and shipment tracking into a single database system. The proposed system improves efficiency, accuracy, and reliability while supporting the smooth functioning of an e-commerce business.